

COWRA, Australia

WMO: 957210

Lat: 33.8450S

Lon: 148.6497E

Elev: 297

StdP: 97.81

Time Zone: 10.00 (AUS)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-1.1	0.0	-4.5	2.7	16.8	-2.8	3.1	12.2	9.2	12.2	8.3	11.6	0.8	150	0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.9	37.8	19.5	35.7	19.2	33.8	18.6	22.1	30.6	21.3	29.6	20.7	28.8	5.3	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	15.5	23.6	19.2	14.5	23.6	18.2	13.6	23.3	66.3	30.2	63.4	29.9	60.9	28.2	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.3	8.2	7.1	-3.1	41.0	1.1	1.9	-3.8	42.4	-4.5	43.5	-5.1	44.5	-5.9	45.9		
			-3.4	23.5	1.1	1.1	-4.2	24.2	-4.9	24.8	-5.5	25.4	-6.3	26.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.2	25.4	23.9	21.0	16.6	11.9	9.3	8.3	9.0	11.8	15.8	19.4	22.3
	DBStd	6.61	3.48	3.20	3.25	3.08	2.66	2.43	2.17	2.20	2.63	3.58	3.63	3.44
	HDD10.0	173	0	0	0	1	12	41	63	46	9	1	0	0
	HDD18.3	1483	1	1	13	68	199	271	313	290	196	97	28	7
	CDD10.0	2428	477	389	340	198	71	21	9	15	64	179	282	382
	CDD18.3	694	219	157	95	14	0	0	0	0	1	17	61	130
	CDH23.3	8783	2806	1790	989	223	4	0	0	0	38	348	893	1691
	CDH26.7	4176	1557	885	386	52	0	0	0	0	4	102	374	818
Wind	WSAvg	2.9	3.4	3.2	2.9	2.5	2.2	2.4	2.5	2.6	2.9	3.0	3.3	3.4
Precipitation	PrecAvg	595	64	49	41	45	49	41	53	51	54	59	50	47
	PrecMax	965	232	169	163	278	166	97	155	102	131	125	115	195
	PrecMin	217	1	2	1	0	1	3	8	4	10	0	6	0
	PrecStd	181	58	45	35	54	39	23	33	26	31	36	31	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.8	39.7	35.0	31.3	23.8	19.3	18.4	21.2	26.9	32.1	36.7	39.0
		MCWB	20.3	20.7	19.0	17.3	14.9	12.4	12.7	13.7	13.6	16.4	18.0	18.5
	2%	DB	38.2	36.8	32.6	27.9	21.8	17.0	16.3	18.5	24.2	29.5	33.3	35.7
		MCWB	19.7	19.7	18.1	16.2	14.1	11.9	11.4	11.9	13.9	16.0	17.2	18.9
	5%	DB	36.3	34.0	30.7	25.7	20.1	15.7	14.9	16.9	21.7	27.2	31.0	33.5
		MCWB	19.7	18.8	18.0	15.6	13.5	11.5	10.7	11.2	13.1	15.3	16.9	18.1
	10%	DB	34.2	31.9	28.8	24.0	18.3	14.4	13.5	15.2	19.6	25.0	28.4	31.3
		MCWB	19.4	18.7	17.7	14.9	12.6	11.1	9.8	10.4	12.4	14.9	16.2	17.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.2	23.3	21.6	19.1	16.9	14.0	15.0	14.6	16.7	19.1	21.0	22.2
		MCDB	31.9	30.1	27.7	24.7	21.1	15.6	16.2	18.7	22.5	24.9	28.2	28.8
	2%	WB	22.2	22.1	20.6	17.9	15.2	13.0	12.6	13.0	15.3	18.0	19.8	21.0
		MCDB	31.7	30.3	26.9	23.8	19.8	15.3	14.9	17.0	20.7	24.8	27.1	28.3
	5%	WB	21.5	21.2	19.7	16.9	14.1	12.2	11.2	11.8	14.2	17.0	18.8	20.0
		MCDB	31.0	28.5	26.3	23.5	18.9	15.0	13.8	15.6	19.6	24.0	26.2	28.0
	10%	WB	20.8	20.4	18.8	15.9	13.1	11.4	10.1	10.7	13.1	16.0	17.8	19.1
		MCDB	30.2	27.9	25.7	22.5	17.6	14.1	13.0	14.8	18.2	22.9	25.0	27.4
Mean Daily Temperature Range	5% DB	MDBR	14.9	13.9	13.3	14.1	12.9	10.1	10.8	12.3	14.6	15.7	15.0	15.1
		MCDBR	17.9	17.7	16.4	16.8	15.3	12.2	13.4	15.6	19.4	19.4	18.8	18.5
		MCWBR	5.3	5.6	6.0	7.3	8.4	7.5	8.7	9.4	10.2	8.4	6.5	5.8
	5% WB	MCDBR	14.1	12.3	12.5	13.5	13.3	10.3	11.2	13.4	15.3	15.4	13.7	13.6
		MCWBR	4.9	4.7	5.6	6.4	7.8	6.7	7.8	8.8	9.1	8.0	5.9	5.5
Clear-Sky Solar Irradiance	taub	0.350	0.346	0.325	0.304	0.291	0.284	0.279	0.287	0.317	0.330	0.345	0.339	
	taud	2.487	2.528	2.555	2.584	2.575	2.609	2.613	2.573	2.473	2.458	2.466	2.506	
	Ebn at Noon	987	970	955	923	883	866	890	929	948	975	986	1001	
	Edh at Noon	116	107	97	83	74	67	70	83	103	113	117	114	
All-Sky Solar Radiation	RadAvg	7.57	6.54	5.53	4.31	3.12	2.36	2.65	3.56	4.99	6.24	6.92	7.63	
	RadStd	0.42	0.50	0.42	0.32	0.20	0.21	0.17	0.28	0.41	0.50	0.58	0.37	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (24 neighbors)	+0.45	N/A	N/A	+1.03	N/A	N/A	N/A	-69	+154	+82

Nomenclature: See separate page